

Modeling Magnetization Transfer for Biological-like Systems Using a Semi-solid Pool with a Super-Lorentzian Lineshape and Dipolar Reservoir

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Magnetization transfer is modeled as an exchange between a liquid pool and a semi-solid pool where the semi-solid pool has, in addition to the Zeeman reservoir, a dipolar reservoir. The lineshape for the liquid pool is characterized by the usual Lorentzian, whereas the semi-solid pool is characterized as a super-Lorentzian. Three systems were investigated: (1) a membrane mixture of a phosphatidylcholine lipid (POPC) and cholesterol; (2) lyophilized white matter at eight solid concentrations ranging from 1 to 24%; and (3) fresh white matter. In all systems, this model fitted the experimental data well with the effect of the dipolar reservoir being most important for the membrane mixture. For the tissue, the dipolar reservoir is not required and the two-pool model with a single Zeeman reservoir for the semi-solid spins characterizes the experimental data equally well.

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INTRODUCTION

Magnetization transfer has been an area of research in biomedical NMR since it was popularized in the paper by Wolff and Balaban in 1989 (1) discussing spin exchange between water and semi-solid protons in tissues *in vivo*. Recently a number of clinical applications of this relatively new imaging technique for obtaining contrast in tissues have emerged (2–9). However, the process of magnetization transfer in complicated systems such as tissues is not yet fully understood. A two-pool exchange model has been shown to successfully characterize magnetization transfer in a model system consisting of agar gel (10). Recently, a three-pool model has been successful at modeling cross-linked bovine serum albumin (11) and, with the addition of a dipolar reservoir for the semi-solid pool, calf lens nuclear homogenate (12). This three-pool model is relatively complicated and requires 10 free parameters, as opposed to only 5 parameters in the two-pool model. Only one additional parameter is required for the dipolar order. Since it is desirable to keep the model as simple as possible and to minimize the number of variables in a fit, we decided to explore the applicability of the two-pool model with and without dipolar effects, to biological tissue-like systems and mammalian tissue.

The initial two-pool model used for investigating agar (13), lipid bilayers (14), rat muscle (15), calf lens homogenate (16), and heat-denatured egg albumin (17) was based on the modified Bloch equations for magnetization transfer, which are identical to the formulation originally presented by McConnell for chemical exchange (18). This model provides an adequate interpretation for experiments involving a single off-resonance irradiation power but does not describe magnetization-transfer data as a function of both irradiation power and offset frequency. However, a slightly modified two-pool model was later found to successfully explain agar gel (10) due to the observation by Swanson (19) that the Bloch equation assumption of a Lorentzian lineshape for the semi-solid pool was incorrect. Instead, a Gaussian lineshape characterizes spins which are homogeneously coupled to one another through the homonuclear dipole–dipole Hamiltonian. This Gaussian semi-solid two-pool model was used to interpret agar gel experiments at several offset frequencies and powers and at various gel concentrations (10).

Based on the success of this model in agar, we have investigated several other systems which are somewhat more complicated and biologically relevant: (1) a membrane mixture composed of two major components of white matter lipids: phosphatidylcholine (PC) and cholesterol (20); (2) reconstituted lyophilized bovine white matter at a range of solid concentrations; and (3) fresh bovine white matter. For the membranes, we chose to model the magnetization transfer as exchange between two pools and to determine the lineshape that characterizes the semi-solid spins in this system. The best lineshape for this system was super-Lorentzian, the lineshape that is expected to arise from partially ordered materials such as biological membranes or molecules adsorbed on surfaces (21). However the two-pool model with this lineshape did not quite capture the magnetization-transfer data and thus we extended our model to include dipolar relaxation of the semi-solid pool (12, 22). With the inclusion of the dipolar relaxation into the two-pool model and a super-Lorentzian semi-solid lineshape, we were able to characterize the magnetization-transfer data for the model membrane systems very well. In lyophilized bovine white matter, we

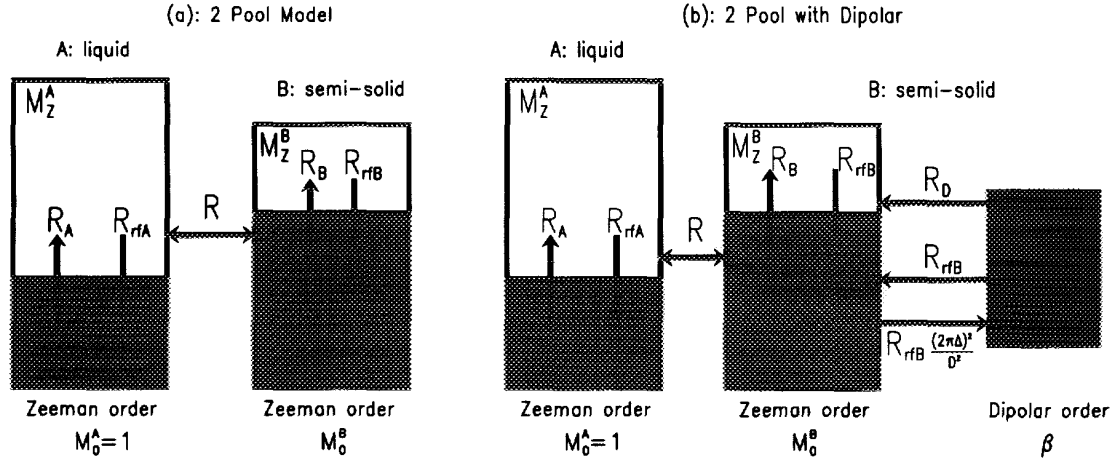


FIG. 1. Two-pool model for magnetization transfer without (a) and with (b) dipolar order included. All rates and populations are defined in the text. The shaded regions represent spins which are not aligned longitudinally.

investigated the concentration dependence of magnetization transfer and showed that the same single model accounts for a range of concentrations of the semi-solid pool. Increasing the concentration of the semi-solid only increases the magnetization exchange and adjusts the T_1/T_2 ratio of the liquid pool but leaves all other parameters of the model unchanged. These results increase confidence that the fitted model is actually characterizing the underlying biophysics of the white matter system and is not just accidentally providing an acceptable fit. Finally, magnetization-transfer experiments performed on excised bovine white matter show very similar behavior and confirm the validity of the same model in *ex vivo* tissue.

THEORY

Two-Pool Model

The two-pool model of magnetization transfer is shown in Fig. 1a. There are two pools: a liquid (pool A) and a semi-solid (pool B). Each pool has an independent longitudinal relaxation rate ($R = 1/T_1$), R_A and R_B . R_{rfA} and R_{rfB} are the rates at which longitudinal magnetization is lost due to the direct effects of the off-resonance irradiation:

$$R_{rfi}(2\pi\Delta) = \omega_1^2 \pi g_i(2\pi\Delta), \quad [1]$$

where $g_i(2\pi\Delta)$ is the absorption lineshape for the spins in pool i , Δ is the offset frequency of the irradiation in hertz, and ω_1 is the amplitude of the irradiation in rads ($\omega_1 = \gamma H_1$) (19). By convention, the number of spins in pool A, M_0^A , is normalized to 1. The magnetization-transfer rate between the two pools is characterized by a fundamental rate constant, R . Since the pools remain constant in size, and since the rate of exchange depends on the concentrations of both pool

A and pool B, the rate of exchange in either direction is a second-order rate given by $RM_0^B M_0^A$. (Even though this reduces to RM_0^B through the convention that $M_0^A = 1$, it does not imply a first-order rate constant as has been suggested by others.) In our formulation, R is a rate constant which is independent of the concentration of the semi-solid compartment (10).

The complete set of differential equations describing these processes are discussed in detail in several references

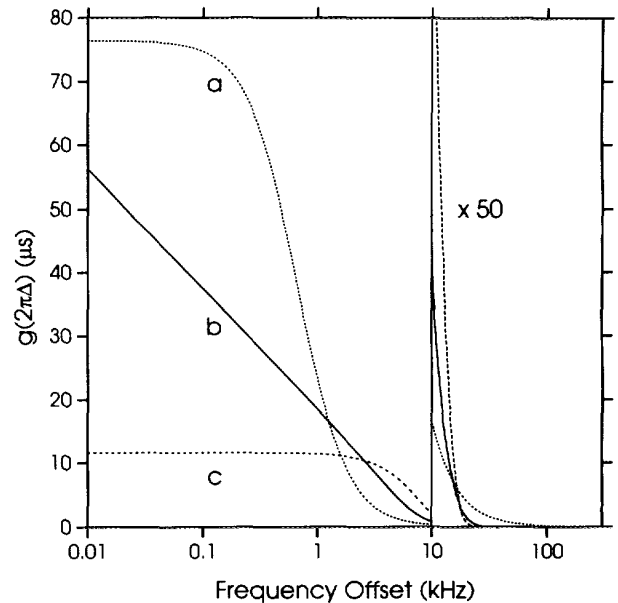


FIG. 2. Lineshapes: (a) Lorentzian (Eq. [5]) with $T_{2B} = 240 \mu s$; (b) super-Lorentzian (Eq. [8]) with $T_{2B} = 17.7 \mu s$; and (c) Gaussian (Eq. [6]) with $T_{2B} = 29 \mu s$. The broad nature of the Lorentzian is highlighted in the magnified lineshapes above 10 kHz. The frequency is plotted on a log axis for consistency with subsequent magnetization-transfer figures.

(10, 15, 23) and therefore, for conciseness, we give only the differential equations for M_Z^A and M_Z^B

$$\begin{aligned}\frac{dM_Z^A}{dt} &= R_A(M_0^A - M_Z^A) - RM_0^B M_Z^A \\ &\quad + RM_0^A M_Z^B - R_{rfA} M_Z^A \\ \frac{dM_Z^B}{dt} &= R_B(M_0^B - M_Z^B) - RM_0^A M_Z^B \\ &\quad + RM_0^B M_Z^A - R_{rfB} M_Z^B.\end{aligned}\quad [2]$$

If the lineshape for the liquid pool is assumed to be Lorentzian and $(2\pi\Delta T_{2A})^2 \gg 1$ for all significantly nonzero points, where $1/T_{2A}$ characterizes the width of the lineshape, then $R_{rfA} = [\omega_1/(2\pi\Delta)]^2(1/T_{2A})$. The solution for the A spins is

$$\begin{aligned}M_Z^A &= \frac{[RM_0^B/R_A]R_B + R_B + R + R_{rfB}}{[RM_0^B/R_A](R_B + R_{rfB}) \\ &\quad + \{1 + (\omega_1/2\pi\Delta)^2[1/R_A T_{2A}]\}(R_B + R_{rfB} + R)} \\ &\equiv \frac{S + R_{rfB}}{T + R_{rfB}U},\end{aligned}\quad [3]$$

where, for later conciseness, we define

$$\begin{aligned}S &= \left[\frac{RM_0^B}{R_A} \right] R_B + R_B + R \\ T &= S + \left(\frac{\omega_1}{2\pi\Delta} \right)^2 \left[\frac{1}{R_A T_{2A}} \right] (R_B + R) \\ U &= \left[\frac{RM_0^B}{R_A} \right] + 1 + \left(\frac{\omega_1}{2\pi\Delta} \right)^2 \left[\frac{1}{R_A T_{2A}} \right].\end{aligned}\quad [4]$$

There are five unique parameters to be determined: R , R_B , $[RM_0^B/R_A]$, $[1/R_A T_{2A}]$, and T_{2B} , (via R_{rfB}). As pointed out by Caines *et al.* (15), we cannot obtain unique model parameters R_A , T_{2A} , R_B , T_{2B} , R , and M_0^B since experiments are performed in the steady state and thus do not contain explicit reference to absolute time scales. If there is no magnetization transfer (i.e., $R = 0$, $RM_0^B = 0$), Eq. [3] reduces to the direct effect, i.e., the effect an off-resonance irradiation pulse has on the liquid pool alone.

Lineshapes

In this paper three lineshapes are presented for characterizing the semi-solid pool: Lorentzian, Gaussian, and super-Lorentzian. We also investigated others such as convolutions of Lorentzians and Gaussians and polynomial expressions, but none proved satisfactory. The Gaussian lineshape is associated with spins in a rigid system where the spin-spin

dipolar interactions are static, whereas the narrow Lorentzian lineshape is associated with spins where these interactions are averaged out due to motions (i.e., liquids). The value of $1/T_2$ characterizes the width of the lineshape. For a Lorentzian lineshape,

$$g(2\pi\Delta) = \frac{T_2}{\pi} \frac{1}{[1 + (2\pi\Delta T_2)^2]} \quad [5]$$

and for a Gaussian

$$g(2\pi\Delta) = \frac{T_2}{\sqrt{2\pi}} \exp\left(-\frac{(2\pi\Delta T_2)^2}{2}\right). \quad [6]$$

These lineshapes are plotted in Fig. 2. The T_2 values in the figure are from fits of the membrane-mixture magnetization-transfer data to the two-pool model and we chose these values in order to illustrate the lineshapes using widths relevant to the work presented in this paper. In particular, we want to point out the T_2 value for the Lorentzian: T_2 has to be quite large (compared to the Gaussian T_2) since a Lorentzian is very broad in the wings (>10 kHz).

As mentioned in the Introduction, the super-Lorentzian lineshape is expected to arise from partially ordered materials. It was first observed in potassium laurate (21) and subsequently in lamellar liquid crystals (24) and lipids (25, 26). The super-Lorentzian is a result of a nonzero time-averaged dipolar Hamiltonian, $\mathcal{H}_d$. In molecules such as lipids, the nonisotropic, planar symmetry of the membrane ensures that this Hamiltonian does not average to zero. The rapid rotation of the molecule about its long axis projects the dipolar interactions along the symmetry axis for the motion, i.e., the normal to the bilayer. The dipolar Hamiltonian scales with the angle θ between the external magnetic field and the bilayer normal (24, 25, 27):

$$\overline{\mathcal{H}_{d,\theta}} = \overline{\mathcal{H}_{d,\theta=0}} \left(\frac{3 \cos^2\theta - 1}{2} \right). \quad [7]$$

If one assumes at $\theta = 0$ the lineshape is Gaussian, then the super-Lorentzian lineshape is given by

$$\begin{aligned}g(2\pi\Delta) &= \int_0^{\pi/2} d\theta \sin\theta \sqrt{\frac{2}{\pi}} \frac{T_2}{|3 \cos^2\theta - 1|} \\ &\quad \times \exp\left(-2\left(\frac{2\pi\Delta T_2}{|3 \cos^2\theta - 1|}\right)^2\right)\end{aligned}\quad [8]$$

and an example is given in Fig. 2. Details of this lineshape can be found in (24, 26, 28).

Two-Pool Model with a Dipolar Reservoir

The two-pool model with the semi-solid pool divided into Zeeman and dipolar order is represented schematically in

Fig. 1b. In the presence of off-resonance irradiation, there is spin exchange between the Zeeman and dipolar order of solids (29, 30). The possibility of having dipolar order in solids arises from the nonzero time-averaged dipolar Hamiltonian, unlike in liquids where the dipolar Hamiltonian is averaged to zero. There are two methods for populating the dipolar reservoir: The Jeener-Brœckært echo sequence (31) and secondly, off-resonance irradiation, as described here. The rate of exchange between Zeeman and dipolar reservoirs, which depends on the amplitude and off-resonance frequency of the irradiation pulse and the lineshape of the semi-solid pool, affects the number of spins in the Zeeman order that can undergo magnetization transfer with the liquid spins and thus needs to be considered in magnetization transfer (12, 22). Depending on how effective the coupling is between the two orders and how quickly the dipolar order relaxes back to Zeeman, it may or may not play an important role in magnetization transfer. A thorough derivation of the dipolar and Zeeman order equations of evolution for a solid under RF irradiation is given by Goldman (30) and has been included in magnetization-transfer models by Yeung *et al.* (22) and Yang and Schleich (12). In this paper we just present the relevant differential equations and the solution for M_Z^A .

The differential equations presented in Eq. [3] modified to include the dipolar reservoir are

$$\begin{aligned} \frac{dM_Z^A}{dt} &= R_A(M_0^A - M_Z^A) - RM_0^B M_Z^A \\ &\quad + RM_0^A M_Z^B - R_{rfA} M_Z^A \\ \frac{dM_Z^B}{dt} &= R_B(M_0^B - M_Z^B) - RM_0^A M_Z^B \\ &\quad + RM_0^B M_Z^A - R_{rfB} M_Z^B + R_{rfB} \beta' \\ \frac{d\beta'}{dt} &= R_{rfB} \left(\frac{2\pi\Delta}{D} \right)^2 (M_Z^B - \beta') - \frac{1}{T_D} \beta' \end{aligned} \quad [9]$$

where $\beta' = \Delta\beta$, β is the inverse spin temperature associated with the dipolar order, T_D is the dipolar relaxation time, and D is related to the NMR linewidth of the semi-solid (30). The solution for M_Z^A is

$$M_Z^A = \frac{S[1 + (2\pi\Delta/D)^2 R_{rfB} T_D] + R_{rfB}}{T[1 + (2\pi\Delta/D)^2 R_{rfB} T_D] + R_{rfB} U} \quad [10]$$

where S , T , and U are defined in Eq. [4] and, once again, the liquid is assumed to be Lorentzian with the width satisfying $(2\pi\Delta T_{2A})^2 \gg 1$.

The additional parameter in this model, T_D , characterizes the relaxation of the dipolar order back to Zeeman order. When the dipolar reservoir relaxes very quickly it essentially plays no role in the model, as expected since the spins are

quickly returned to Zeeman order. Thus when $T_D = 0$, Eq. [10] reduces to Eq. [3]. The rate and degree of coupling between the Zeeman and dipolar orders depends on the line-shape and linewidth of the semi-solid pool and the amplitude and offset frequency of the irradiation pulse. Although the effect of the dipolar reservoir is a reduction in the number of semi-solid spins able to undergo magnetization transfer, adjustment of M_0^B is not appropriate since the number of spins lost to the dipolar order varies with irradiation offset frequency and amplitude.

As mentioned above, D is related to the NMR linewidth of the semi-solid and therefore depends on the lineshape $g(2\pi\Delta)$ chosen for this pool. When the spin-spin interactions are purely dipolar $D^2 = 1/3T_{2B}^2$ for a Gaussian lineshape (30). For the super-Lorentzian, we chose the lineshape at $\theta = 0$ to be a Gaussian characterized by T_{2B} . If we average D^2 over all angles (i.e., $\int_0^{\pi/2} D^2 d\theta \sin \theta$, where $D^2 = 1/(3T_{2B}(\theta)^2)$), then $D^2 = 1/15T_{2B}^2$.

EXPERIMENTAL CONSIDERATIONS

NMR Measurements

To fully explore magnetization transfer, comprehensive experiments involving several offset frequencies and several irradiation powers must be done. All our experiments were performed on a NMR spectrometer (SMIS, Surrey, England) with a 30 cm, horizontal-bore, superconducting

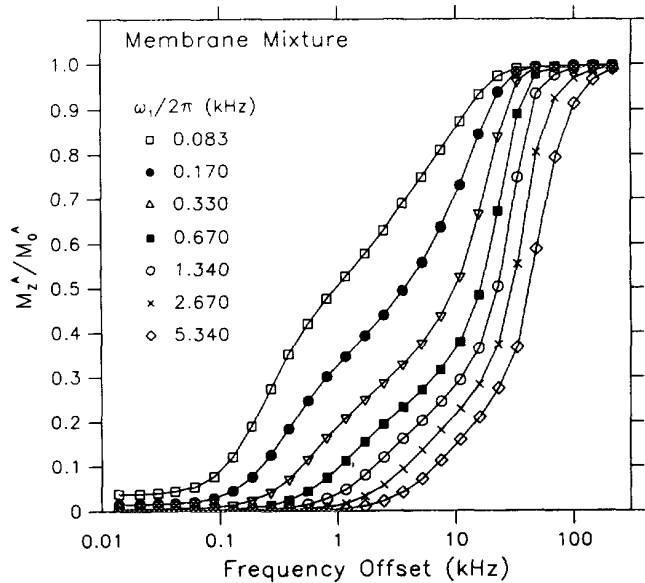


FIG. 3. Magnetization-transfer data for model membranes. Normalized residual longitudinal magnetization (M_Z^A/M_0^A) is plotted versus the frequency offset (Δ) on a logarithmic scale. Data for seven different amplitudes of H_1 are shown: $\omega_1/2\pi = 83$ ($\square$), 170 ($\bullet$), 330 ($\triangle$), 670 ($\blacksquare$), 1340 ($\circ$), 2670 ($\times$), and 5340 ($\diamond$) Hz. Solid lines are to guide the eye.

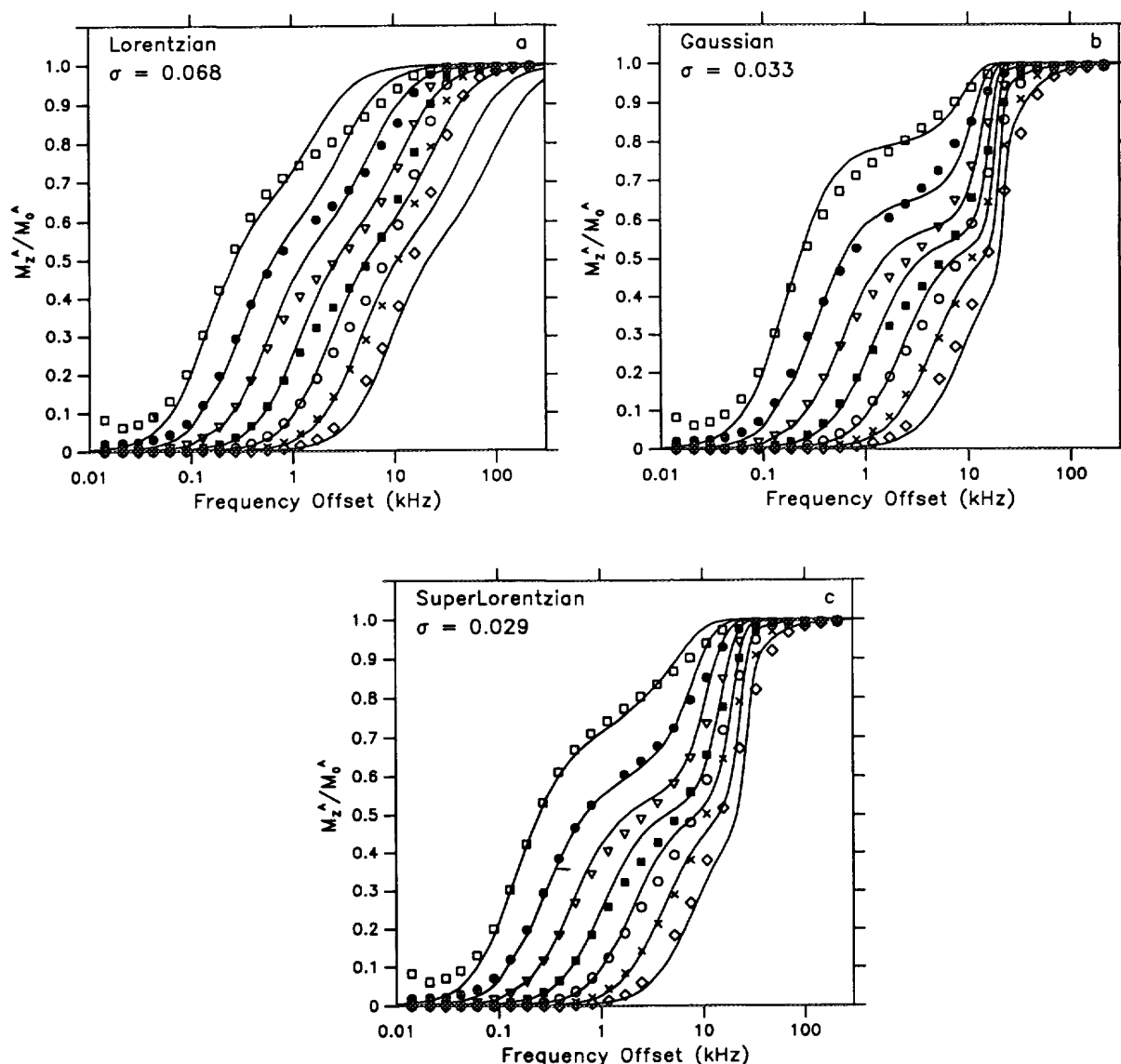


FIG. 4. Magnetization-transfer data for model membranes as in Fig. 2. The solid lines are fits to Eq. [3] with semi-solid lineshapes as follows: (a) Lorentzian, (b) Gaussian, and (c) super-Lorentzian. Average residual deviation per point is given on each graph. Fitting parameters are given in Table 1.

magnet (Nalorac, Martinez, California) operating at 1.5 T. The pulse sequence for the off-resonance irradiation experiment incorporated a 7 s pulse of amplitude H_1 at offset frequency Δ followed by a $40 \mu\text{s}$ 90° pulse 200 μs later. Data acquisition started 20 μs after the 90° pulse. Ten points were sampled with a 12 bit ADC over a 100 μs time and averaged to improve signal to noise. Transverse magnetization following the off-resonance RF was compensated for by phase cycling both the 90° pulse and data acquisition. The 27 different values of Δ were distributed along a logarithmic scale from 0.014 to 213 kHz. The H_1 amplitude was increased in six steps by factors of two from 0.083 to 5.34 kHz and was calibrated by deter-

mining the length of a rectangular pulse required to invert the spin population. The duration of the off-resonance pulse was 7 s to ensure that steady state had been reached (10). The time between data acquisition and the start of the next saturation pulse was set to 5 s.

In the magnetization-transfer experiment, we measure the liquid pool, i.e., the water protons. A sample of our magnetization-transfer data is given in Fig. 3. M_z^A / M_0^A is plotted versus Δ on a logarithmic scale for the seven H_1 amplitudes. The signals were normalized to unity for no off-resonance irradiation. The powers and corresponding symbols are given in the figure. Reproducibility of normalized magnetization-transfer signals was better than

TABLE 1
Parameters of the Fit of Eq. [3] to the Membrane-Mixture Magnetization-Transfer Data
for Three Different Semi-Solid Lineshapes

Lineshape	R (s^{-1})	$\frac{RM_B^0}{R_A}$	$\frac{1}{R_A T_{2A}}$	T_{2B} (μs)	σ
Lorentzian	22 ± 3	0.75 ± 0.04	5.1 ± 0.4	240 ± 30	0.068
Gaussian	20 ± 3	0.84 ± 0.05	5.7 ± 0.5	29.0 ± 1.6	0.033
Super-Lorentzian	24 ± 4	0.97 ± 0.05	4.6 ± 0.5	17.7 ± 1.2	0.029

Note. $R_B = 1 s^{-1}$.

0.5% of full scale. At low offset frequencies (<100 Hz), M_z^A/M_0^A is higher than expected for the 0.083 kHz power due to B_0 inhomogeneity over the sample, and these points are excluded from the fit.

All the fitting was done using the downhill simplex method, a sum of squares minimization routine (32). Errors in the fit are given as the average residual deviation per point, σ , between the fit and experiment ($\sigma = (1/N)(\sum_{i=1}^N [fit(x_i) - data(x_i)]^2)^{1/2}$), where N is the number of data points used in the fit. This is not a χ^2 measurement since we are not interested in determining the parameters of the fit to within experimental error, but rather we are interested in the validity of the structure of the model. The error of a fitted parameter was determined by finding the maximum and minimum values of the parameter, while all other parameters remain fixed, for an increase of 1 in χ^2 .

Samples

The membrane sample was a mixture of 11% (total weight) 1-palmitoyl-2-oleoyl-phosphatidylcholine (POPC) and 7% cholesterol (Avanti Polar Lipids, Alabaster, Alabama). The mixture was made from chloroform stock solutions of each lipid. The two solutions were combined in the appropriate proportions and the chloroform removed under a stream of nitrogen gas, followed by placement under vacuum for several hours. The dried lipid mixture was suspended in buffer (Na_2HPO_4/NaH_2PO_4 , pH 5), gently vortexed above the main transition temperature, and submitted to four cycles of freeze-thawing. We performed our experiments at pH 5, consistent with other published magnetization-transfer studies of lipids (32), but not a physiological pH (20).

The lyophilized white-matter samples were prepared in the following manner. Approximately 12 g of freshly excised bovine white matter was mixed with 50 ml of distilled water, frozen to $-24^\circ C$ and lyophilized for 24 h. As a result of lyophilization, ~ 3 g of dry weight powdered white matter was obtained. Eight different concentrations, 1.0, 2.0, 4.0, 8.0, 12.0, 16.0, 20.0, and $24.0\% \pm 0.2\%$, of white matter were prepared by dissolving 5, 10, 20, 40, 60, 80, 100, and 120 mg, respectively, in the appropriate amount of distilled

water to give a total weight of 0.5 g. A pH of 6.3 ± 0.1 for all samples was measured before and after the magnetization-transfer experiment.

The white-matter sample was taken from a bovine corpus colosum. The experiment was performed within two hours post mortem.

All samples were sealed in test tubes with parafilm. Each set of experiments took approximately two hours. Before and after the magnetization-transfer experiment, the multicomponent T_2 was measured using a CPMG sequence to ensure samples had not degraded or changed appreciably during the course of the experiment. Multicomponent T_2 measurements are most sensitive to tissue change with time after excision. In no case was any degradation in the T_2 spectrum observed. All experiments were done at room temperature (20 – $22^\circ C$).

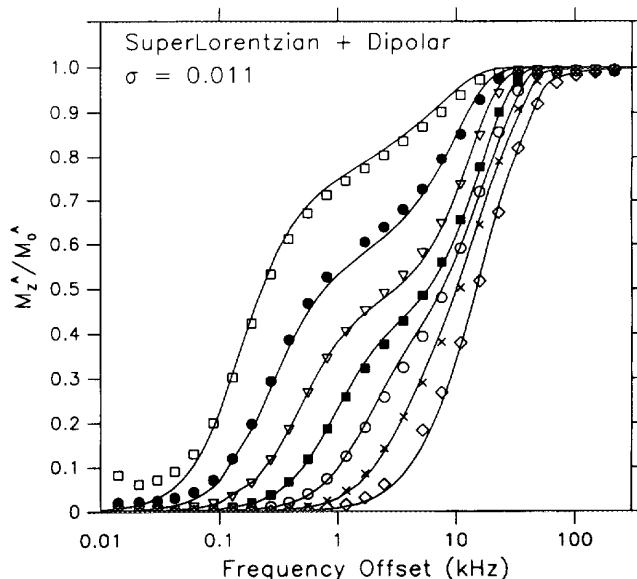


FIG. 5. Magnetization-transfer data for model membranes as in Fig. 2. The solid lines are fits to Eq. [10] with a super-Lorentzian semi-solid lineshape. Fitting parameters are given in Table 2.

TABLE 2
Parameters of the Fit of Eq. [3] (– Dipolar) and Eq. [10] (+ Dipolar) Using a Super-Lorentzian Lineshape

System	Max MT	Super-Lorentzian	R (s^{-1})	$\frac{RM_0^B}{R_A}$	$\frac{1}{R_A T_{2A}}$	T_{2B} (μs)	T_d (ms)	σ
Membranes	0.45	– dipolar	24 ± 4	0.97 ± 0.05	4.6 ± 0.5	17.7 ± 1.2	—	0.029
		+ dipolar	40 ± 4	1.41 ± 0.08	4.4 ± 0.5	10.8 ± 0.7	3.3 ± 0.8	0.011
24% White matter	0.57	– dipolar	38 ± 4	2.1 ± 0.1	12.2 ± 1.7	10.4 ± 0.5	—	0.023
		+ dipolar	52 ± 5	2.7 ± 0.1	11.6 ± 1.7	9.2 ± 0.4	0.8 ± 0.2	0.017
White matter	0.54	– dipolar	21 ± 3	2.0 ± 0.1	23 ± 4	10.4 ± 0.5	—	0.022
		+ dipolar	26 ± 3	2.3 ± 0.1	22 ± 3	9.7 ± 0.4	0.7 ± 0.3	0.019

Note. $R_B = 1 s^{-1}$.

RESULTS AND DISCUSSION

Using the two-pool model for magnetization exchange, the membrane-mixture magnetization-transfer data was fitted to Eq. [3] using various semi-solid lineshapes. Figure 4 demonstrates that none of the three lineshapes, Lorentzian, Gaussian, or super-Lorentzian, is completely satisfactory, but there is a significant improvement in the quality of fit from 4a ($\sigma = 0.068$) to 4c ($\sigma = 0.029$). The parameters for these three fits are given in Table 1. The same data fit to Eq. [10], the two-pool model including dipolar order, is shown in Fig. 5 and the parameters are listed in Table 2. A super-Lorentzian lineshape was used and the fit is very good, $\sigma = 0.011$. We also fitted the same equation using a Gaussian lineshape, but it was worse than that for the super-Lorentzian ($\sigma = 0.020$). A Lorentzian

lineshape is not appropriate for a semi-solid pool with a dipolar reservoir. For all systems, R_B was fixed at $1 s^{-1}$, since if left free tended toward 0. $R_B > 4 s^{-1}$ increased the average residual deviation per point, whereas for $R_B < 4 s^{-1}$ the parameter had no effect. Discussion of the parameters in Tables 1 and 2 is postponed until the results for all systems are presented.

To study the exchange model more rigorously, it is useful to look at the concentration dependence of magnetization transfer. We did this with lyophilized white matter, a more tissue-relevant system than the membrane mixture, and we recognize that, although these samples have a similar composition to intact tissue, the complete structural integrity cannot be assured. The resulting data for a concentration of 24%, which is the solid concentration in white matter, is presented in Fig. 6 and fitted using a model with (6b) and

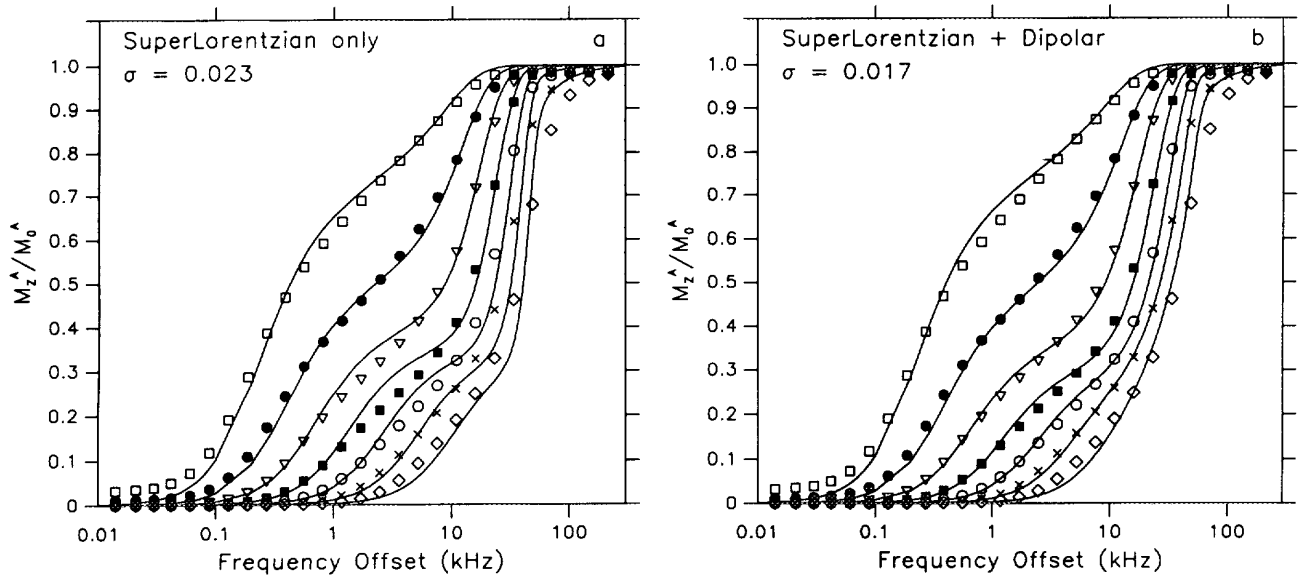


FIG. 6. Magnetization-transfer data for 24% lyophilized white matter. Same experimental procedure as in Fig. 2. Solid lines are fits to (a) Eq. [3] (no dipolar reservoir) and (b) Eq. [10] (with dipolar reservoir) with a semi-solid super-Lorentzian lineshape. Fitting parameters are given in Table 2.

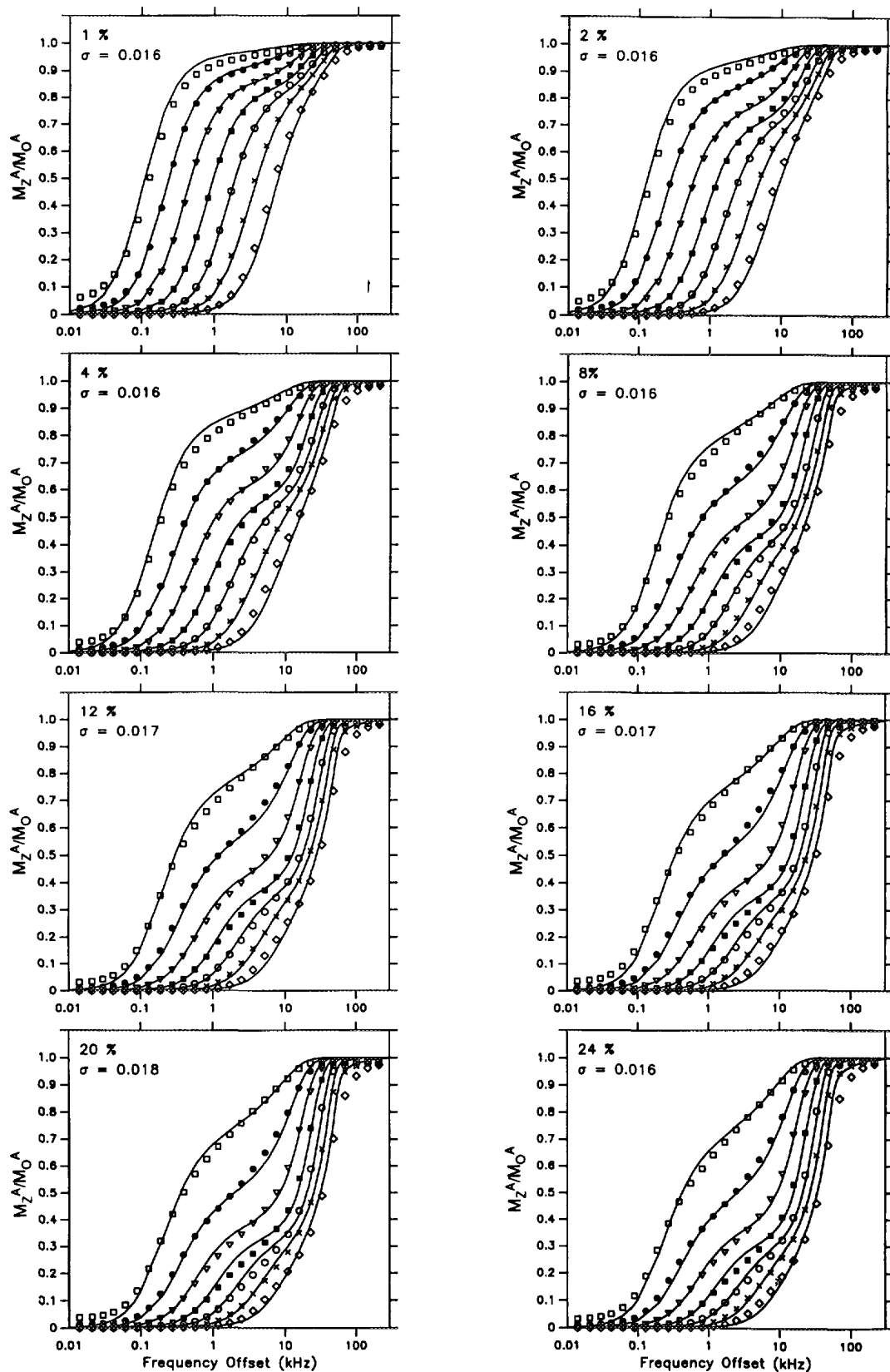


FIG. 7. Magnetization-transfer data for lyophilized white matter at various concentrations. Solid lines are the global fit of Eq. [10]. The fit is described in the text and parameters for the fit are given in Table 3.

TABLE 3

Max MT and Concentration-Dependent Parameters for the Global Fit of Lyophilized White Matter for the Two-Pool Model with a Dipolar Reservoir (Eq. [10])

Concentration (%)	Max MT	$\frac{RM_0^B}{R_A}$	$\frac{1}{R_A T_{2A}}$	σ
1	0.15	0.23 ± 0.03	1.7 ± 0.1	0.016
2	0.24	0.44 ± 0.04	2.1 ± 0.2	0.016
4	0.37	0.87 ± 0.10	3.5 ± 0.4	0.016
8	0.48	1.46 ± 0.08	5.8 ± 0.6	0.016
12	0.52	1.9 ± 0.1	7.4 ± 0.9	0.017
16	0.55	2.1 ± 0.1	8.7 ± 1.1	0.017
20	0.56	2.28 ± 0.11	9.8 ± 1.3	0.018
24	0.57	2.51 ± 0.13	11.9 ± 1.6	0.017

Note. $R_B = 1 \text{ s}^{-1}$, $R = 46 \pm 5 \text{ s}^{-1}$, $T_{2B} = 9.3 \pm 0.5 \mu\text{s}$, and $T_D = 0.8 \pm 0.2 \text{ ms}$. Allowing T_{2B} and T_D to vary with concentration did not significantly improve the fits. Fitting each concentration independently gave the same values of T_{2B} and T_D as the global fit, within experimental error.

without (6a) dipolar order. For both models, the super-Lorentzian gave the best fit. The parameters associated with these two fits are given in Table 2. Once again there is a substantial improvement in the fit with the inclusion of the dipolar order, particularly where the magnetization-transfer effect is greatest ($\Delta = 1\text{--}10 \text{ kHz}$, $\omega_1/2\pi = 0.3\text{--}3 \text{ kHz}$). However the improvement is less striking than for the membrane mixture. White matter is a much more complicated system than the POPC and cholesterol mixture and contains a variety of additional molecules that could be involved in magnetization exchange, such as proteins and other lipids.

The model does not quite capture M_z^A/M_0^A at high offset

frequencies for the high power. This could probably be accounted for by making a small adjustment in the lineshape tail ($>70 \text{ kHz}$), but this would have minimal impact on the fitted parameters and magnetization-transfer model validity. Thus we decided to work with the well-characterized super-Lorentzian.

We measured magnetization transfer of lyophilized white matter at eight concentrations (Fig. 7). The solid lines are a global fit of the two-pool model with dipolar and a super-Lorentzian lineshape. There are four global parameters: R_B , R , T_{2B} , and T_D . The parameters $[RM_0^B/R_A]$ and $[1/R_A T_{2A}]$ were determined separately for each concentration. T_{2B} characterizes the linewidth of the solid and does not depend on concentration. Thus there was a total of 20 parameters to be fitted using eight concentrations ($8 \times 27 \times 7 = 1512$ data points in total). R_B was fixed at 1 s^{-1} for reasons described above. The global parameters are $R = 46 \pm 2 \text{ s}^{-1}$, $T_{2B} = 9.3 \pm 0.5 \mu\text{s}$, and $T_D = 0.8 \pm 0.2 \text{ ms}$ and Table 3 lists the other two parameters for each concentration. The average residual deviation per point, σ , for the total fit was 0.016. On a single concentration the resulting σ was between 0.016 and 0.018, only slightly higher than individually fitting all five parameters at a single concentration. The value of $[RM_0^B/R_A]$ reflects the magnitude of magnetization transfer: what determines the amount of magnetization transfer is the exchange rate RM_0^B relative to the longitudinal relaxation rate R_A . The increase in this parameter, with concentration, is consistent with the increase in magnetization transfer exhibited by the data (see Table 3). Provided the exchange sites M_0^B are not inaccessible, magnetization transfer increases with concentration. (i.e., relaxation rates are expected to increase linearly with concentration, $R_A = C_0 + kM_0^B$,

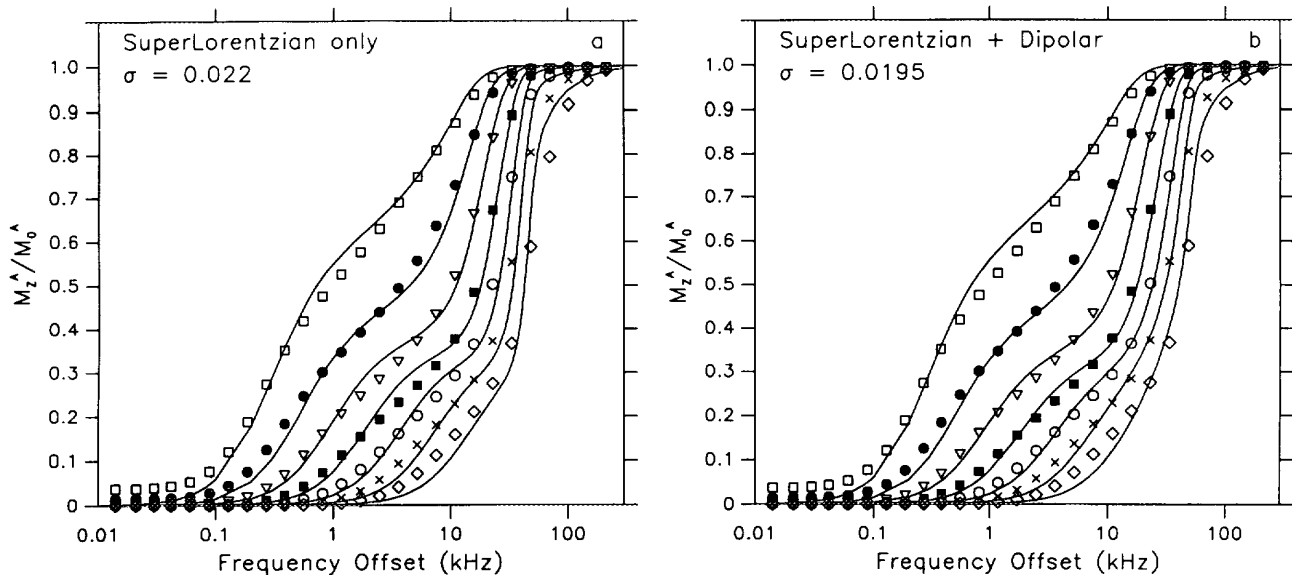


FIG. 8. Magnetization-transfer data for bovine white matter. Solid lines are fits to (a) Eq. [3] (no dipolar reservoir) and (b) Eq. [10] (with dipolar reservoir). The semi-solid lineshape is Lorentzian. Fitted parameters are given in Table 2.

where C_0 is a concentration independent relaxation mechanism and k is a proportionality constant. Provided $C_0 \neq 0$, $[RM_0^B/R_A]$ increases as M_0^B increases.) As indicated in Table 3, eventually max MT will reach a plateau, and increasing the concentration will no longer increase the amount of magnetization transfer. The acceptable fits at all concentrations with global constraints on four of the six parameters are a rigorous test for the consistency and hence validity of this model.

We measured magnetization transfer in fresh bovine white matter as a final sample. The data and best fits are in Fig. 8 and the parameters are in Table 2. Once again a super-Lorentzian lineshape was used and the two-pool model with and without dipolar order. The improvement obtained by including the dipolar reservoir (σ decreases from 0.022 to 0.019) is not as dramatic as in the above systems, indicating that the need for the dipolar reservoir is lost in real tissues. The strong dipolar component we had for POPC and cholesterol no longer plays as important a role, possibly indicating that other molecules which have faster dipolar relaxation are more dominant in the magnetization exchange.

The ratio of T_1 to T_2 ($1/R_A T_{2A}$) increases with system composition complexity (i.e., in real tissue, this parameter is five times as large as in the membrane mixture). This parameter does not change much with the addition of the dipolar reservoir, which is not surprising, given that $[1/R_A T_{2A}]$ is the direct effect contribution. Because of the multicomponent relaxation behavior of tissues and membranes, it is not yet clear how to separate R_A , T_{2A} , and RM_0^B using independent measurements of T_{2A} and R_A via CPMG and inversion recovery sequences, respectively (34).

Our semi-solid T_{2B} , as determined by the fit, is longest in the membrane mixture, indicating a less rigid semi-solid component in the membranes than in the tissue. For comparison, we give some literature values for the second moment M_2 associated with $\mathcal{H}_D$ (27). Several groups prefer to characterize the linewidth using M_2 , where the larger the M_2 value, the more rigid the system on the NMR timescale. For a Gaussian lineshape, $M_2 = 1/T_{2B}^2$, and for the super-Lorentzian, the orientationally averaged M_2 is $1/(5T_{2B}^2)$. Thus, for the membranes, $M_2 = 1.7 \times 10^9 \pm 0.2 \times 10^9 \text{ s}^{-2}$, and, for the white matter, $M_2 = 2.4 \times 10^9 \pm 0.2 \times 10^9 \text{ s}^{-2}$. Bloom *et al.* (26) report M_2 values of $0.4 \times 10^9 \text{ s}^{-2}$ for the phospholipid DPPC just above the gel to liquid-crystalline phase transition (41°C) and $2.5 \times 10^9 \text{ s}^{-2}$ just below. We note that in lipid and cholesterol mixtures with high content of cholesterol, as our sample has, the second moment is much larger than in the lipid system without cholesterol (35). For the integral membrane protein rhodopsin incorporated into DMPC phospholipid bilayers, Bloom *et al.* report an M_2 of $3 \times 10^9 \text{ s}^{-2}$. An even more rigid system is wood where $M_2 = 5.6 \times 10^9 \text{ s}^{-2}$ (36).

Our fits give a T_D of $3.3 \pm 0.8 \text{ ms}$ for the membranes, $0.8 \pm 0.2 \text{ ms}$ for 24% lyophilized white matter, and 0.7 ± 0.3

ms for white matter. For comparison, MacKay *et al.* (37) measured a T_D of 7 ms in bean cell walls ($M_2 = 3 \times 10^9 \text{ s}^{-2}$) and Jeffrey *et al.* (38) found $T_D = 200 \text{ ms}$ in a soap-water system just above the gel to liquid-crystalline system ($M_2 = 3 \times 10^8 \text{ s}^{-2}$) which increased with increasing temperature. Both experiments were done at 90 MHz. Our short T_D results are consistent with our large M_2 values (i.e., short T_{2B}) and the shorter T_D in the tissue sample indicates a more rigid semi-solid than in the membranes.

Tables 2 and 3 also list the max MT for each system. Max MT is defined as the maximum difference between the direct effect and M_Z^A/M_0^A . As can be seen, the trends in max MT and $[RM_0^B/R_A]$, the parameter which represents the magnitude of magnetization transfer, are parallel, giving further credibility to the model. It is RM_0^B relative to R_A which determines the amount of magnetization transfer.

Due to the complexity of Eqs. [3] and [10], the consequences for the model parameters of adding the additional parameter T_D are not obvious. In all three systems, as can be seen from the parameters given in Table 2, T_{2B} decreases, whereas R and $[RM_0^B/R_A]$ increase with the inclusion of the dipolar reservoir. The most significant change in parameters occurs for the membrane mixture, the system with the largest T_D and the most dramatic improvement with the inclusion of the dipolar reservoir. For white matter, the changes are essentially within experimental error.

CONCLUDING REMARKS

The two-pool model with the inclusion of a dipolar reservoir for the semi-solid spins fits experimental magnetization-transfer data for a membrane mixture of POPC and cholesterol very well. We looked at several lineshapes for the semi-solid pool and found that a super-Lorentzian characterized it the best. The same model, with different parameters, also fits lyophilized white matter well over a range of solid concentrations. For the third system we examined, fresh white matter, the dipolar reservoir is not as dominant in the two-pool model as it was in the membranes and the parameters (other than T_D) are essentially, within experimental error, the same with and without the dipolar reservoir. Similarly, for fresh white matter a super-Lorentzian lineshape characterizes the semi-solid lineshape. From these results we believe we have an appropriate and relatively simple model for the magnetization-transfer processes in tissue.

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